

Oil Price Shocks and Texas Energy Employment

Executive Brief — Interactive Dashboard Companion

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1. Research Question

Do changes in West Texas Intermediate (WTI) oil prices predict county-level employment growth in Texas energy-related industries (NAICS 211 oil and gas extraction; NAICS 213 mining support), and did this relationship change after the 2020 oil-price shock?

This brief is the executive companion to an interactive dashboard (app.py) that lets a stakeholder explore the same panel — 9,295 county-quarter observations across 202 Texas counties from 2010 Q2 through 2025 Q3 — under selectable counties, time windows, and pre/post-2020 splits.

2. Motivation

Texas is the largest oil-producing state in the United States, and major price disruptions in 2014 and 2020 are widely associated with regional layoffs, capital-spending cuts, and credit stress for energy-services firms. The static narrative in macro commentary is that WTI movements drive Texas energy employment in lockstep. Using the team's county-quarter panel, we test that assumption and find a more nuanced picture: the statewide relationship is statistically present but economically small, and the response varies sharply by county and by sample period.

3. Headline Findings

Three facts emerge from the panel and from the dashboard's live OLS recomputation across slices:

- **Statewide elasticity is small.** A \$1 increase in WTI is associated with ~0.17 pp higher quarterly energy-employment growth across Texas ($\beta = +0.0017$, $p < 0.01$). Oil prices alone explain less than 1% of the variance in county-level employment growth at the panel level.
- **Sensitivity roughly doubles post-2020.** The same coefficient rises from +0.0010 (pre-March 2020, not significant) to +0.0035 (post-March 2020, $p < 0.01$). The pandemic and subsequent rate cycle did not break the oil-employment link; it tightened it.
- **County heterogeneity is the dominant story.** Harris (Houston) and Ector (Odessa) show R^2 of 0.36 and 0.21 respectively — meaningful explanatory power for individual counties. Midland, despite being the most oil-exposed county by name recognition, shows R^2 of only 0.01 because its energy employment kept rising through both shocks. The aggregate hides a 30x range in county-level fit.

4. Selected Slices from the Live Dashboard

Slice	β (WTI)	p-value	R^2	N	Read
All Texas, full window	+0.0017	0.001	0.001	9,295	Statewide signal is weak
All Texas, pre-2020	+0.0010	0.077	0.001	5,911	Pre-pandemic: not significant
All Texas, post-2020	+0.0035	0.003	0.003	3,384	Magnitude doubles after shock

Slice	β (WTI)	p-value	R^2	N	Read
Harris (Houston), full	+0.0010	<0.001	0.36	62	Big-county signal is clean
Ector (Odessa), full	+0.0016	<0.001	0.21	62	Classic Permian-basin response
Midland, full	+0.0007	0.43	0.01	62	Decoupled — see brief discussion

All values are produced live by the dashboard's OLS panel and replicate the team's analysis notebook (03_analysis.ipynb).

5. How to Use the Dashboard

The companion application is a single-file Plotly Dash app. Three controls drive every selection:

1. **County dropdown** — switch between the panel-wide “All Texas” average and the top 12 Texas counties by mean energy employment (Harris, Midland, Ector, Tarrant, Dallas, etc.). Selecting a single county filters the regression to that county's 60+ quarterly observations.
2. **Sample-split dropdown** — restrict the regression to pre-2020 (Jan 2010–Feb 2020), post-2020 (Mar 2020–present), or the full window. Combined with the slider, this is how you reproduce the structural-break analysis in row 2–3 of the table above.
3. **Time-window slider** — a two-handle range slider for any year span from 2010 to 2025. Useful for isolating individual shocks (e.g., 2014–2016 oil crash) or stable periods (e.g., 2017–2019 recovery).

The chart and the OLS coefficient panel below it both recompute on every selection. The R^2 updates in real time, making it easy to see how much of the variation in employment growth is captured by oil prices in any given slice.

To launch: open a terminal in the project folder and run “python app.py”. The app loads at <http://127.0.0.1:8050>. Full instructions and dependency pins are in `DASHBOARD_README.md` and `requirements.txt`.

6. Limitations

- **Contemporaneous specification.** The OLS uses same-quarter WTI. Hiring decisions almost certainly respond with a lag of one to several quarters. A distributed-lag specification would likely raise R^2 and shift coefficient magnitudes.
- **Omitted variables.** Rig counts (Baker Hughes), Permian production volumes, capex announcements, and firm-level wage data are not included. Their absence likely understates the structural component of energy-employment movements.
- **Aggregation across counties.** The “All Texas” view is an unweighted average of county growth rates. This treats Loving County (population ~50) the same as Harris County (population 4.7M). The dashboard mitigates this by letting users isolate individual counties.
- **Association, not causation.** WTI prices respond to the same global macro forces that drive U.S. labor demand. A genuine causal estimate would require an instrument (e.g., OPEC supply shocks).

7. Implications for Practitioners

Audience	Key Takeaway
Energy-company FP&A	Do not size hiring plans off WTI alone. The aggregate elasticity is small (~1.7 pp employment growth per \$10 oil move) and post-shock heterogeneity across counties is large.
Regional bank credit analysts	Houston and Permian-basin counties (Ector) show strong, predictable WTI sensitivity with $R^2 > 0.20$. Use county-specific models for energy-sector loan exposure rather than statewide aggregates.
State policymakers	Texas's energy-employment exposure to oil prices roughly doubled post-2020, even though the absolute magnitude remains small. Workforce-development spending should account for the structural shift, not just the level.
Long-only equity / asset managers	WTI is a weak headline signal for Texas-based energy-services equities at the panel level. For directional bets, county-level or rig-count data is a stronger predictor than statewide averages.

8. Data Sources

- Federal Reserve Economic Data — WTISPLC (WTI Spot Price), TXUR (Texas Unemployment Rate). Federal Reserve Bank of St. Louis.
- Bureau of Labor Statistics — Quarterly Census of Employment and Wages (QCEW), NAICS 211 and 213, Texas counties, 2010 Q2 – 2025 Q3.
- Team repository: github.com/LoganAverill/BANA-4373-Final-Project---Bauer-Daniel-Averill